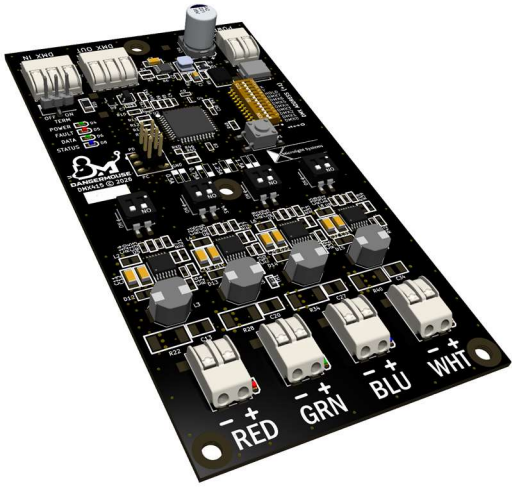
DMX415

DESIGNED & ASSEMBLED
IN THE USA
with imported electronic components

Four channel DMX controlled LED driver



General description

The DMX415 controls four LED driver channels from four consecutive DMX channel values. Each channel has independent PWM brightness control, with square-law brightness mapping and smoothing during normal operation. Local switches select the starting DMX address, stored-output HOLD operation or one of three standalone TEST modes. Each channel has settings for four output currents, from 0.5A to 1.5A.

Protections include thermal, under-voltage, reverse-polarity, and output short circuit.

The four outputs can be assigned to red, green, blue and white LED loads or other compatible LED loads.

Quick start

1. Mount the board to a suitable metal back panel or heat sink using a qualified thermal interface material. Confirm electrical clearance, polarity and connector assignments.

2. With power removed, connect compatible LED loads and select the appropriate channel current settings.

3. For an initial standalone check, set TEST bit 0 (“Y”) OFF and TEST bit 1 (“X”) ON. Apply power: all four outputs should run at approximately 10% PWM duty if voltage and temperature checks permit.

4. For DMX operation, set both TEST switches (X and Y) OFF and set the starting address. All address switches OFF selects address 1. Connect the controller and command channels 1 through 4 (assuming all DIP switches are OFF).

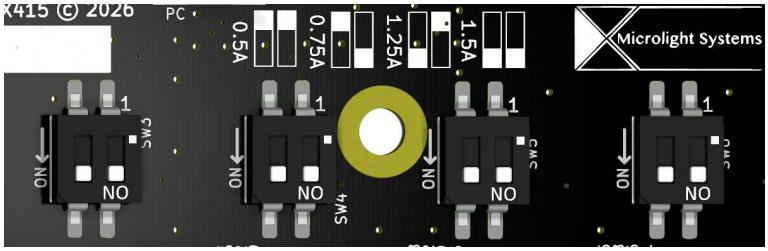
5. After changing switches while powered, press and release the pushbutton to re-scan the switches. In normal mode with HOLD enabled, this also saves the four channel values.

DIP SWITCH SETTINGS

Output Current

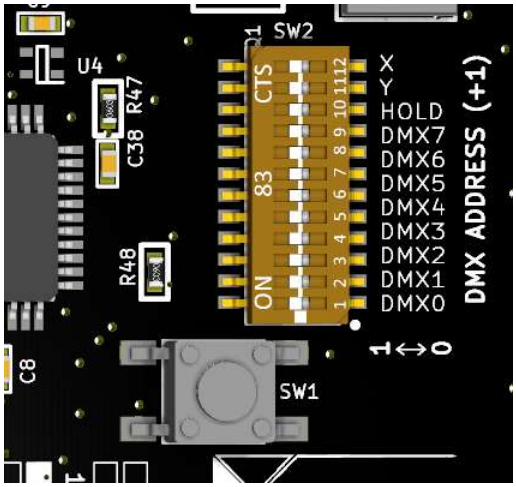
Each channel uses a 2-position DIP switch to set the maximum LED current. Refer to the settings on the board as shown below.

For example, both DIP switches UP = 0.5A output, both DIP switches DOWN = 1.5A maximum output.



DMX Channels

The 12-position DIP switch is used to set the starting DMX channel, the HOLD function, and the TEST modes.



Switch Position(s)	Function
1-8 (labeled DMX0 through DMX7)	Sets the starting DMX address. Add 1 to the binary value of the switches. For example, all switches OFF = DMX Channel 1. For a starting address of 100, DIP switches should be set for decimal 99, or binary 01100011 (shown DMX7 – DMX0)
9 HOLD	When OFF, Dimming values are held for approximately 3 seconds after DMX signal is lost, then turned OFF. When ON, Dimming values are held indefinitely if the DMX signal is lost. **Save Function:** When ON, pressing the switch SW1 will save the current DMX dimming values to non-volatile memory. After a power cycle, if HOLD is still ON, and if there is no DMX data received, the DMX415 will restore the previously saved values. If / when DMX data is received, the stored values are abandoned and the DMX415 resumes normal operation, following the DMX dimming commands.
10, 11 TEST Modes	"X" and "Y" can be used to force test modes, ignoring the DMX input data. This is intended for testing an installation without a DMX source connected.

X	Y	Function
0	0	Normal operation, responds only to DMX input or restored from Memory (see HOLD function above)
0	1	LED outputs are ramped up to full brightness, then down, each channel in sequence.
1	0	All four output channels at 10% brightness continuously
1	1	All four output channels at 100% brightness continuously

Features

Function	Implementation
DMX channels	4 consecutive channels; 8-bit commands
PWM	15-bit PWM for LED Dimming; nominal 980 Hz LED PWM rate
Standalone tests	Sequential fades, all channels at 10% duty, all channels at full duty
Local controls	9 address bits, HOLD, 2 TEST bits and pushbutton
Indicators	POWER, FAULT, DATA and STATUS LED indicators at each output for local indication
Stored output	Four DMX values retained in EEPROM when requested (HOLD activated, press pushbutton to store current dimming values)
Protections	Thermal, Reverse Polarity, Output short circuit

Electrical and environmental specifications

Unless stated otherwise: Vin = 48V, Vout = 20V (x4 channels), Iout = 1.5A, room temperature

Parameter	Min	Typical	Max	Unit
Recommended DC input voltage	22	--	52	V
Absolute maximum (continuous) DC input voltage	--	--	57	V
Recommended External Fuse Vin = 24V Vin = 48V	 -- --	 10A Fast 6A Fast	 -- --	 A A
Input transient voltage and duration	--	--	75 / 10	V / ms
Input current at full load (Vin = 48V, 4x 20V Outputs at 1.5A each)	--	2.7	3	A
Idle input current Vin = 24V Vin = 48V	 -- --	 8 7	 -- --	 mA mA
Channel current accuracy	-8	0	+8	%
Minimum driver headroom (Vin – VLED) Iout = 0.5A Iout = 1.5A	 -- --	 4 5	 6 7	 V V
Efficiency at rated operating point	88	92	--	%
Output ripple current	--	10mA	--	mA p-p
Ambient operating temperature	-20	--	+50	˚C
Storage temperature	-20	--	+65	˚C
DMX termination resistance (when selected by jumper)	-1%	121	+1%	Ω

Control and operating characteristics

Parameter	Value	Condition or qualification
DMX receiver configuration	250 kbit/s; 8 data bits; 2 stop bits; no parity	Configured USART; no formal DMX compliance claim
Starting address	1 to 512 selectable	Use 1 to 509 to fit all four channels in one 512-slot universe
Address encoding	A = 1 + switch-weight sum	Weights 1, 2, 4, 8, 16, 32, 64, 128, 256
Brightness command	0 to 255 per channel	Gamma corrected to 15bit value
PWM frequency	976.56 Hz nominal	Clock tolerance not qualified
TEST mode 2 duty	10% nominal	All four outputs commanded 10% Duty
TEST mode 3 duty	100%	All four outputs commanded fully on
Voltage fault thresholds	Below 22 V / above 70 V	Software thresholds; not hardware ratings
Onboard Thermal Fault >60°C >75°C	 LED Power(s) drop LEDs fully OFF	LED power approximately 1/16 of normal power
Thermal Fault Hysterisis	Approx 10°C	Applies to both fault conditions
Voltage and temperature update	About 0.41 s nominal	For reference only
DMX loss timeout with HOLD off	About 3.1 s nominal	Based on absence of serial receiver activity

TEST selection

Mode	DIP SWITCH “Y”	DIP SWITCH “X”	Output behavior
0	OFF	OFF	Normal DMX control
1	ON	OFF	Fade channels 1 through 4 sequentially
2	OFF	ON	All four channels at 10% PWM duty
3	ON	ON	All four channels at 100% PWM duty

Settings are applied at startup or pushbutton release. TEST operation does not require DMX. In every TEST mode, the thermal fault are fully active.

HOLD and stored output

With HOLD enabled in normal mode, the previous output is indefinitely retained when the DMX signal is lost. Pressing and releasing the pushbutton requests an EEPROM save of the four current DMX command values. On a later startup with HOLD enabled, valid saved values are restored. These saved values are preserved even if power is lost. This feature can be used to program a fixture for a specific color, and it will always operate in that condition as long as it powered up with HOLD activated.
If the DMX signal returns, the DMX415 resumes normal operation.

With HOLD disabled: if DMX is lost, the output(s) shutdown after the normal delay.

Thermal management

Use a good thermal interface material between the board and a metal back panel or heat sink. A large metal panel can spread heat; a suitable heat sink and airflow may be needed for higher output power, high ambient temperature or enclosed installations. Qualify cooling with all four channels loaded simultaneously at the worst intended input voltage and LED load.

The onboard temperature protection may not protect the circuit in all situations and should not be relied on for thermal management.

Onboard Temperature sensor

The onboard temperature sensor is used to indicate potentially damaging high temperature. If the board temperature approaches 60°C, the LEDs will be dimmed to approximately 1/16 of the programmed dim values, to reduce added heat. If the board gets close to 75°C, the LEDs will be turned OFF until the board cools approximately 10°C below the fault thresholds.

Operating mode	Response above the indicated 60°C threshold
Normal DMX with receiver activity	FAULT blinks slowly.
TEST modes 1 2 and 3	FAULT blinks slowly and all four outputs are dimmed. Automatic recovery occurs after the temperature drops approximately 10°C
Normal HOLD without receiver activity	FAULT can blink, but a hot flag does not unconditionally force the held output off in the current implementation.

Hot spot caution

The onboard sensor measures temperature near its physical location. If the board is not properly cooled (heat spreader and/or air flow), hot spots can develop and component failure is possible. Mounting, TIM coverage, and airflow affect the thermistor temperature and hot spot heating. Consequently, the actual component temperature and allowable operating load at which the thermal fault appears will vary by installation and should be verified.

Recommended Thermal Interface Material (TIM)

Henkel BERGQUIST BOND PLY TBP 850, formerly BOND-PLY 100 and commonly identified by BP100 part numbers, is recommended. It is a double-sided thermally conductive adhesive tape. Select thickness to match the mounting surfaces; Henkel lists 5, 8, and 11 mil thicknesses. 11 mil is recommended for most installations. This is considered a permanent bond material. If a removable material is desired, consider a standard gap-pad type material.
Because the back of the PC board has signals and power traces, an insulated (reinforced) material is recommended.

Use a clean, flat surfaces and apply the manufacturer’s recommended bonding process. Thin tape requires close contact and is not a substitute for a gap pad across uneven or protruding surfaces. Confirm adhesion and thermal performance in the intended enclosure.

Indicators and troubleshooting

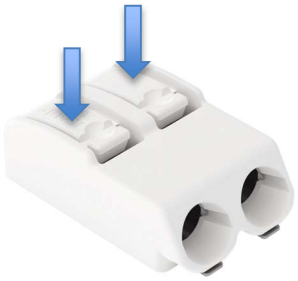
Indicator	Normal indication	Other behavior
POWER (green)	Solid for measured 22–70 V inclusive	Below 22 V: slow blink. Above 70 V: fast blink.
FAULT (red)	Off at or below indicated 60°C	Above 60°C: slow blink.
DATA (amber)	Toggles after a complete set of four addressed channels	Can look steady at high frame rates (fast flashing); retains its last state if reception stops.
STATUS (blue)	On when serial receiver activity was observed	Off when activity stops

Startup runs a five-pass POWER → FAULT → DATA → STATUS → DATA → FAULT chase, nominally 1.5 s total.

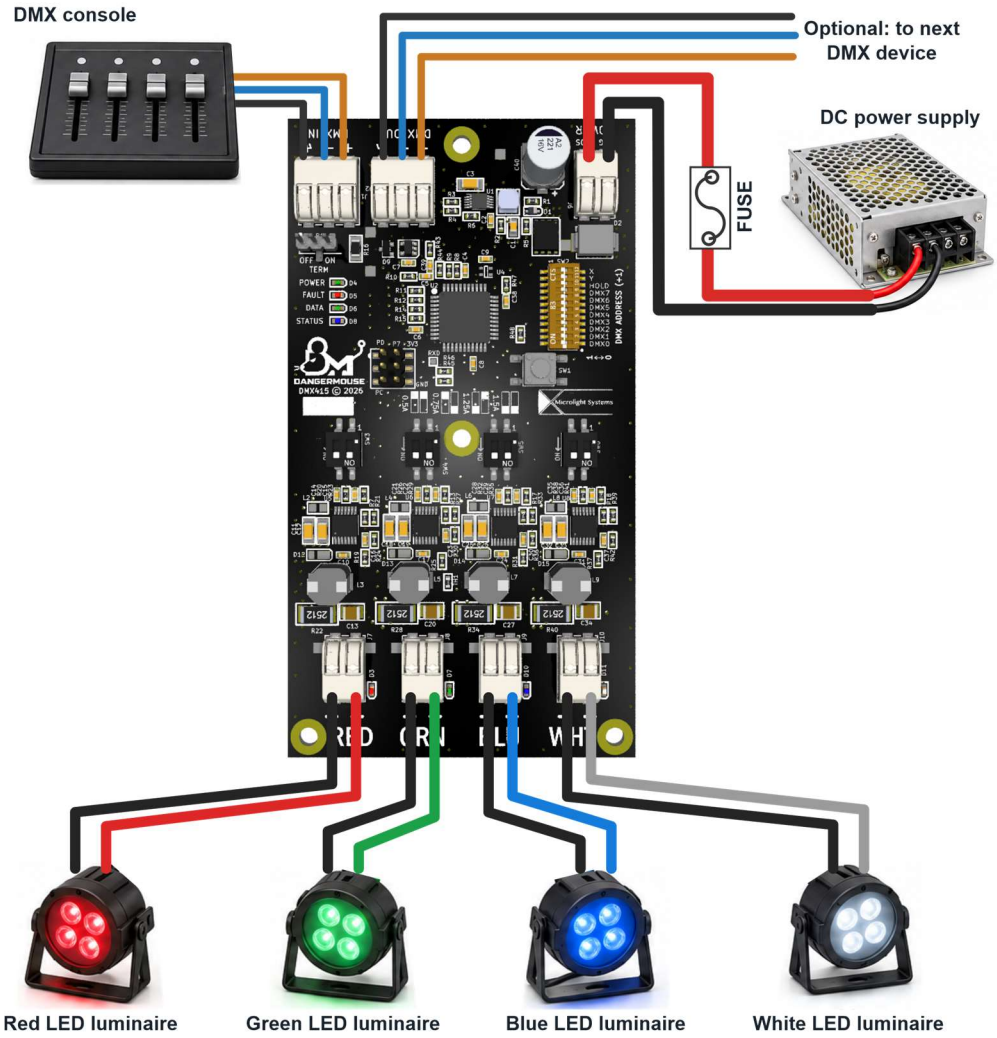
Connections

All connectors (DC Power Input, DMX Input / Output, and Four LED Outputs) use Wago spring-lock (push-in) connectors. The specified wire gauge allowed is 18 – 26AWG. Solid (or stranded + tinned) wire can be pushed in. To release the wire, press on the button on top of the connector.

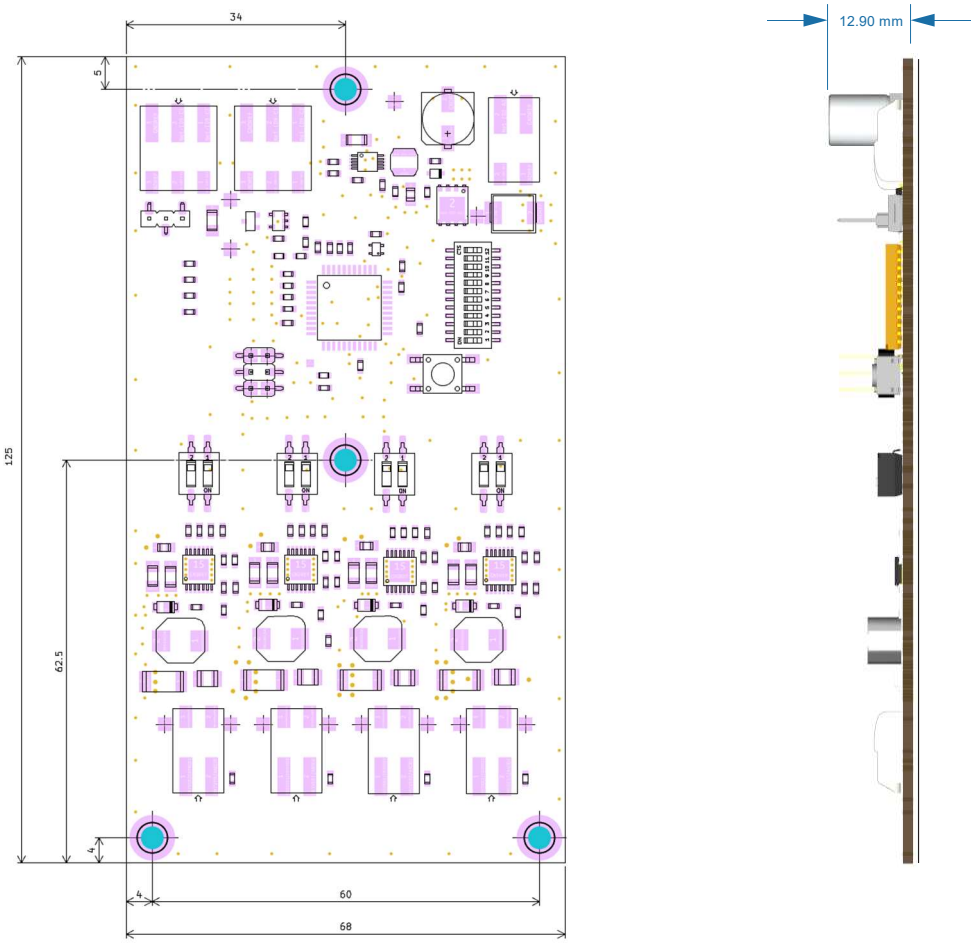
Stranded wire (untinned) can possibly be used, by pressing the button while inserting the wire.



DMX415 Connections



Dimensions



Dimensions in mm.
Mounting Hole Diameter (5x): 3.6mm (0.141”)

Thermal Interface Materials

[1] Henkel Technical Data Sheet BERGQUIST BOND PLY TBP 850, August 2021, formerly BERGQUIST BOND-PLY 100.
https://datasheets.tdx.henkel.com/BERGQUIST-BOND-PLY-TBP-850-en_GL.pdf

[2] Laird Gap Filling thermal pad, available in various thicknesses.
https://datasheets.tdx.henkel.com/BERGQUIST-BOND-PLY-TBP-850-en_GL.pdf